

GENEMIND SAMPLES BATCH – 3 (MAY) 15 SAMPLES COMPARISSION

GENEMIND BATCH 15 SAMPLES : (TOTAL SIZE = 68 GB)

STEP	INPUT FILE	PROCESS	TOOLS USED	DURATION OF TIME	DATA SIZE
1	FASTQ (R1, R2)	MULTIQC ANALYSIS	MULTIQC, FASTQC	2 HRS	14 MB
2	FASTQ (R1, R2)	FASTP ANALYSIS	FASTP	1 HRS	55 GB
3	TRIMMED FASTQ (R1, R2)	MULTIQC ANALYSIS	MULTIQC, FASTQC	2 HRS	14 MB
4	TRIMMED FASTQ (R1, R2)	HS METRICS ANALYSIS	SENTIEON	4 HRS	60 GB
5	TRIMMED FASTQ (R1, R2)	SECONDARY ANALYSIS	SENTIEON	5 HRS	80 GB
6	BAM FILES	CHROMOSOME COVERAGE	INDEXCOV, PANDA	1 HRS	10 GB
7	VCF, BAM AND BED	SAMPLE SEX AND COVERAGE ANALYSIS	VARSEQ VS CLINICAL COVERAGE AND SAMPLE STASTICS	3 DAYS	6 GB
8	VCF	VARAINT DEPTH ANALYSIS	VARSEQ	1 DAYS	20 GB
9	BAM AND VCF	TERTIARY ANALYSIS AND REPORT PREPARATION	VARSEQ	3 DAYS	10 GB
TOTAL				~ 8 DAYS	~ 400 GB

STEP : 1 – MULTIQC ANALYSIS BEFORE TRIMMING

- INPUT FILES : R1, R2 FASTQ FILES
- TOOL : FASTQC, MULTIQC
- TIME TAKEN: (2 HRS FOR 15 SAMPLES)
- DATA SIZE : 14 MB (FINAL OUTPUT)

FASTQC RESULT BEFORE TRIMMING:

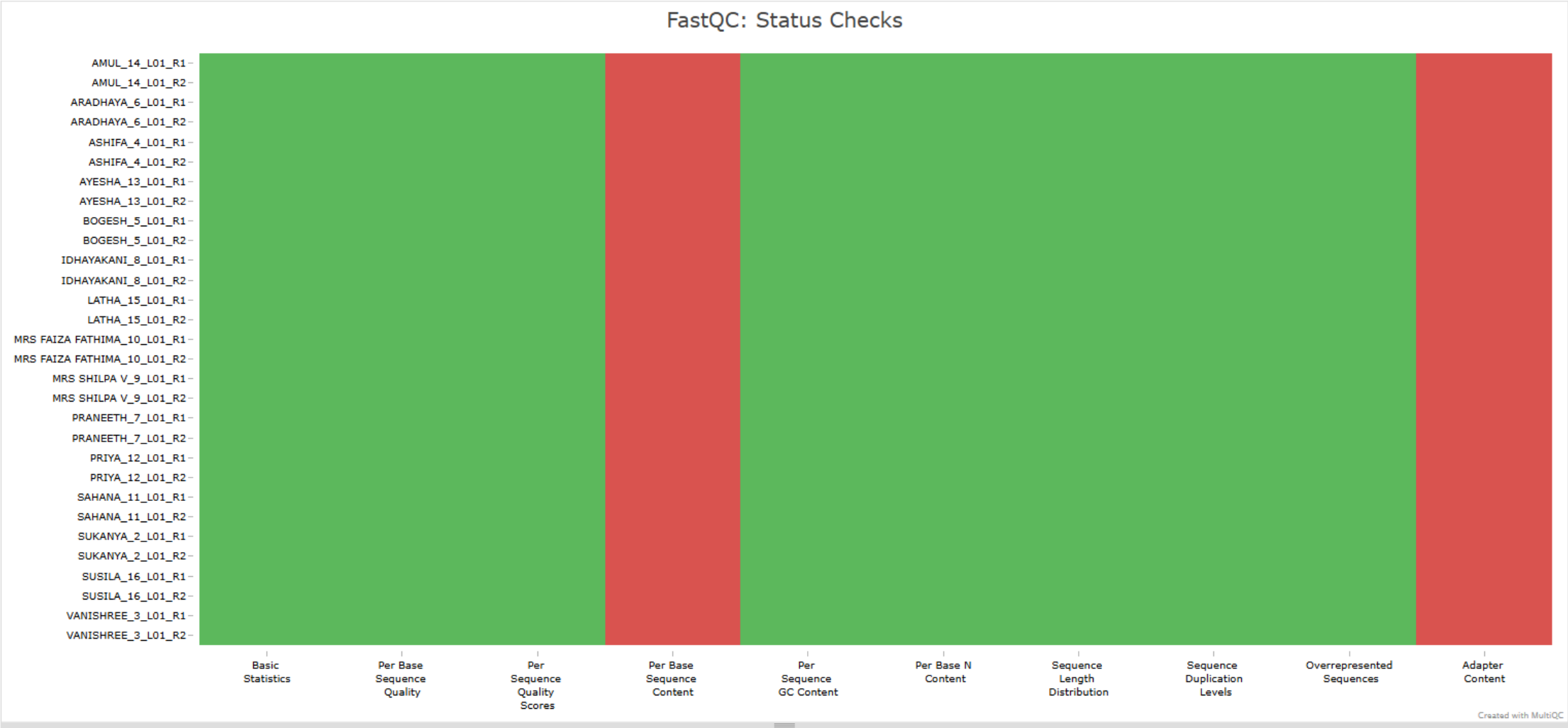
Status Checks

Help

Status for each FastQC section showing whether results seem entirely normal (green), slightly abnormal (orange) or very unusual (red).

Min: Max:

Export Plot



STEP 2 : FASTP (ADAPTER TRIMMING OF FASTQ FILES)

- INPUT FILES : R1, R2 FASTQ FILES
- TOOL : FASTP
- TIME TAKEN: (1 HRS FOR 15 SAMPLES)
- DATA SIZE : 55 GB (FINAL OUTPUT)

FASTQC RESULT:

ADAPTER TRIMMING OF FASTQ FILES

BEFORE AND AFTER FASTP READS COMPARISSION:

GENEMIND - BATCH -3 MAY							
BEFORE FASTP				AFTER FASTP			
Sample_ID	Sample_Name (Sample Names may mismatched)	NUMBER OF READS	SIZE OF READS	TOTAL DATA GENERATED	NUMBER OF READS	SIZE OF READS	TOTAL DATA GENERATED
2	SUKANYA – Agilent CREV4	46.861958 M	150bp	7.029294 G	46.681348 M	142bp	6.645241 G
3	VANISHREE – Agilent CREV4	53.528012 M	150bp	8.029202 G	53.359170 M	142bp	7.591863 G
4	ASHIFA -Agilent CREV4	48.754322 M	150bp	7.313148 G	48.574088 M	142bp	7.591863 G
5	BOGESH - Agilent CREV4	45.193576 M	150bp	6.779036 G	45.041430 M	144bp	6.499689 G
6	ARADHAYA - Agilent CREV4	52.382534 M	150bp	7.857380 G	52.211876 M	142bp	7.415593 G
7	PRANEETH – Agilent CREV4	47.112950 M	150bp	7.066942 G	46.941252 M	141bp	6.653597 G
8	IDHAYAKANI – Agilent CREV4	53.523662 M	150bp	8.028549 G	53.329946 M	141bp	7.569533 G
9	MRS SHILPA V – Agilent V8	66.874416 M	150bp	10.031162 G	66.656100 M	136bp	9.080622 G
10	MRS FAIZA FATHIMA Agilent V8	72.677694 M	150bp	10.901654 G	72.449488 M	136bp	9.867102 G
11	SAHANA – Agilent V8	62.170240 M	150bp	9.325536 G	61.992792 M	133bp	8.272118 G
12	PRIYA – Agilent V8	59.523224 M	150bp	8.928484 G	59.354740 M	135bp	8.026278 G
13	AYESHA – Agilent V8	60.245158 M	150bp	9.036774 G	60.037138 M	139bp	8.390268 G
14	AMUL - Agilent V8	69.318586 M	150bp	10.397788 G	69.100306 M	139bp	9.408717 G
15	LATHA – Agilent V8	61.119628 M	150bp	9.167944 G	60.912356 M	136bp	8.338588 G
16	SUSILA – Agilent V8	58.793466 M	150bp	8.819020 G	58.561652 M	137bp	8.057449 G

STEP : 3 – MULTIQC ANALYSIS AFTER TRIMMING

- INPUT FILES : TRIMMED R1, R2 FASTQ FILES
- TOOL : FASTQC, MULTIQC
- TIME TAKEN: (2 HRS FOR 15 SAMPLES)
- DATA SIZE : 14 MB (FINAL OUTPUT)

FASTQC RESULT AFTER TRIMMING:

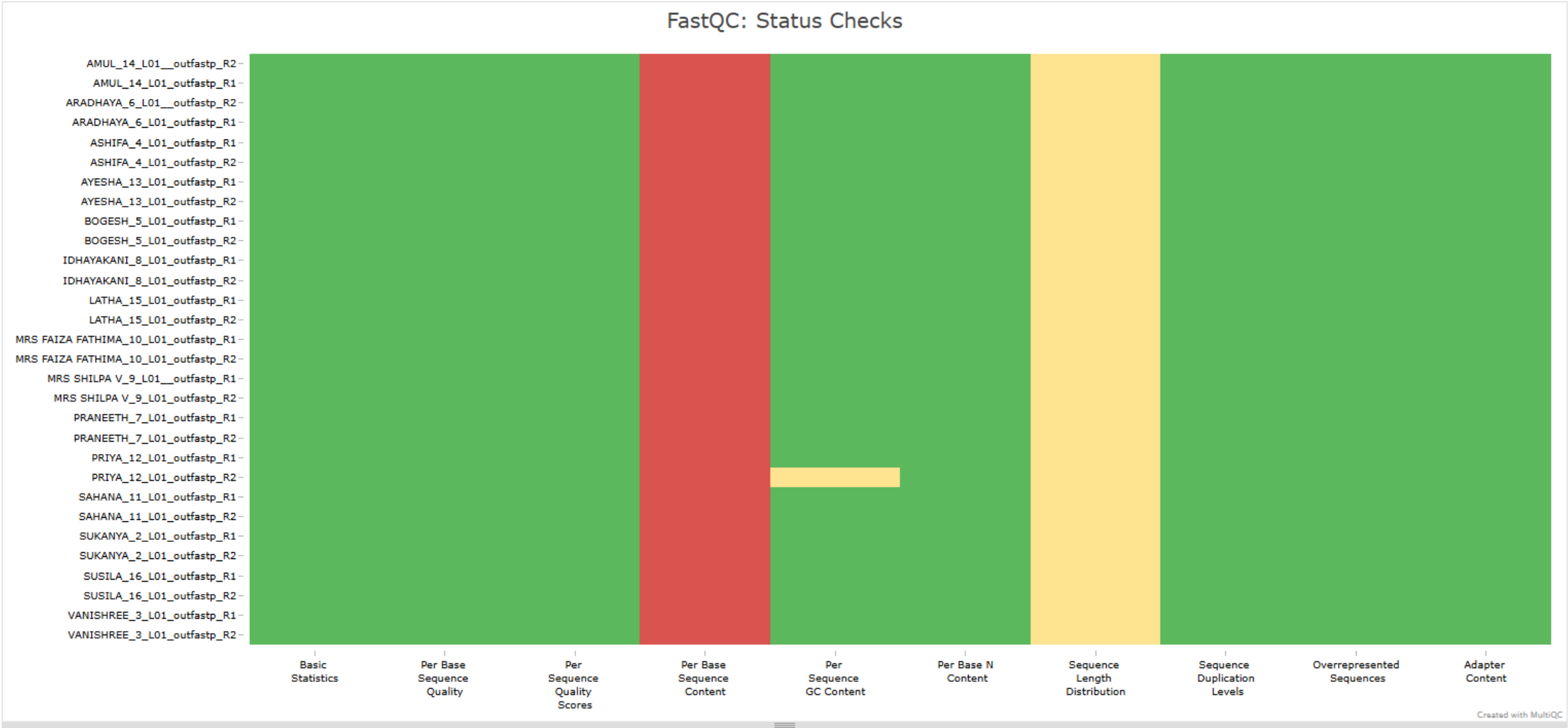
Status Checks

Status for each FastQC section showing whether results seem entirely normal (green), slightly abnormal (orange) or very unusual (red).

Min: Max:

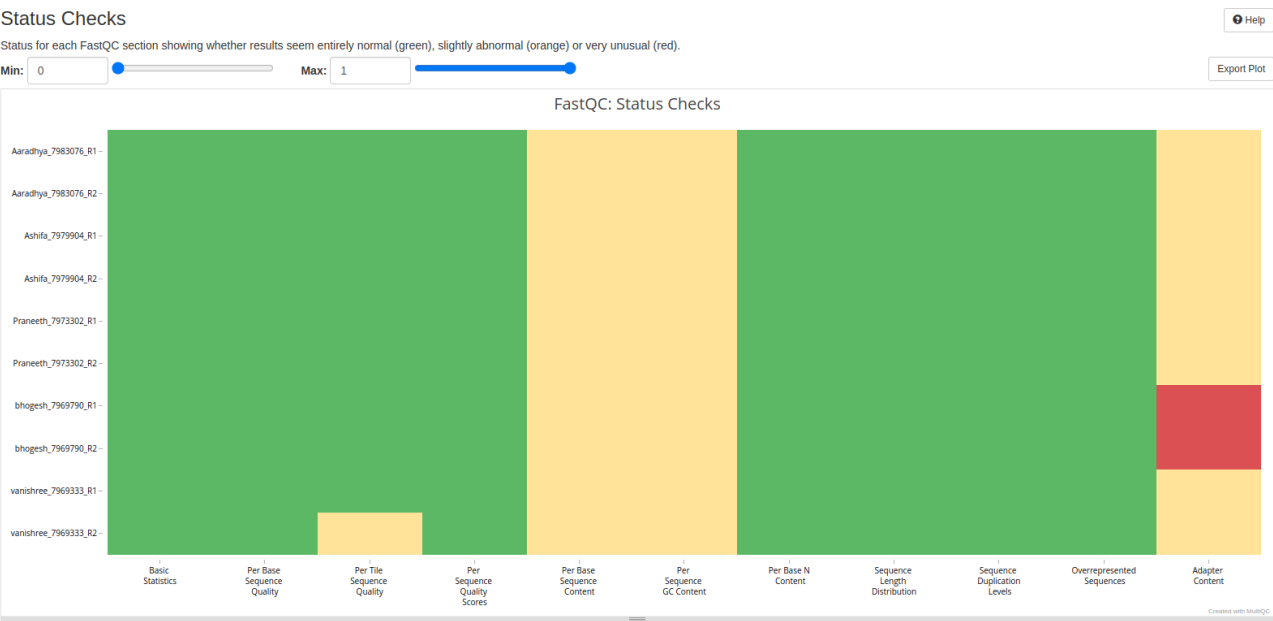
Help

Export Plot



MEDGENOME AND GENEMIND COMPARSSION FASTQC RESULT BEFORE TRIMMING:

MEDGENOME (BEFORE FASTP):



GENEMIND (BEFORE FASTP):



STEP 4 : HS METRICS ANALYSIS :

- INPUT FILES : TRIMMED R1, R2 FASTQ FILES
- TOOL : SENTIEON
- TIME TAKEN: (4 HRS FOR 15 SAMPLES)
- DATA SIZE : 60 GB (FINAL OUTPUT)

RESULT: HS METRICS (SureSelectXT_Clinical_Research_Exome_V4 BED):

MIN_MAP=15, MIN_BASE_QUAL_20, COVERAGE_CAP_1000							
BAIT_SET		BAIT					
	AGILENT_CREV4_BED						
NAME	SUKANYA	VANISHREE	ASHIFA	BOGESH	ARADHAYA	PRANEETH	IDHAYAKANI
GENOME_SIZE	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917
BAIT_TERRITORY	51149367	51149367	51149367	51149367	51149367	51149367	51149367
TARGET_TERRITORY	41089482	41089482	41089482	41089482	41089482	41089482	41089482
BAIT_DESIGN_EFFICIENCY	0.803323	0.803323	0.803323	0.803323	0.803323	0.803323	0.803323
TOTAL_READS	46681348	53359170	48574088	45041430	52211876	46941252	53329946
PF_READS	46681348	53359170	48574088	45041430	52211876	46941252	53329946
PF_UNIQUE_READS	46681348	53359170	48574088	45041430	52211876	46941252	53329946
PCT_PF_READS	1	1	1	1	1	1	1
PCT_PF_UQ_READS	1	1	1	1	1	1	1
PF_UQ_READS_ALIGNED	46679436	53357073	48572398	45039874	52210036	46939218	53327695
PCT_PF_UQ_READS_ALIGNED	0.999959	0.999961	0.999965	0.999965	0.999965	0.999957	0.999958
PF_BASES_ALIGNED	6398709566	7311423413	6651463026	6270803674	7140842318	6403590663	7284933804
PF_UQ_BASES_ALIGNED	6398709566	7311423413	6651463026	6270803674	7140842318	6403590663	7284933804
ON_BAIT_BASES	3486243413	4059632085	3780755952	3400140699	3945999612	3579186251	4109123295
NEAR_BAIT_BASES	1141428512	1300502396	1192945579	1120134969	1282160104	1127152304	1275743590
OFF_BAIT_BASES	1771037641	1951288932	1677761495	1750528006	1912682602	1697252108	1900066919
ON_TARGET_BASES	2185104784	2589177381	2374861399	2118238771	2498300936	2250950858	2575673015
PCT_SELECTED_BASES	0.72322	0.733118	0.747761	0.720845	0.732149	0.734953	0.739179
PCT_OFF_BAIT	0.27678	0.266882	0.252239	0.279155	0.267851	0.265047	0.260821
ON_BAIT_VS_SELECTED	0.753347	0.757375	0.760149	0.752198	0.754759	0.760503	0.763087
MEAN_BAIT_COVERAGE	68.1581	79.3682	73.916	66.4747	77.1466	69.9752	80.3358
MEAN_TARGET_COVERAGE	53.1792	63.0131	57.7973	51.5518	60.8015	54.7817	62.6845
PCT_USABLE_BASES_ON_BAIT	0.524623	0.534735	0.547195	0.523124	0.532122	0.537933	0.54285
PCT_USABLE_BASES_ON_TARGET	0.328822	0.341046	0.343717	0.325898	0.336898	0.338306	0.340268
FOLD_ENRICHMENT	34.2707	34.9255	35.7535	34.106	34.7588	35.1575	35.4798
ZERO_CVG_TARGETS_PCT	0.0165546	0.0168916	0.0167347	0.0172379	0.0158252	0.0167439	0.0160884
PCT_EXC_DUPE	0	0	0	0	0	0	0
PCT_EXC_MAPQ	0.0490881	0.0450476	0.0472243	0.0489914	0.0478962	0.0490253	0.0478593
PCT_EXC_BASEQ	0.00939723	0.00704596	0.00957339	0.00830463	0.00767492	0.00952489	0.00920778
PCT_EXC_OVERLAP	0.292459	0.293885	0.299631	0.289063	0.293335	0.301011	0.305239
PCT_EXC_OFF_TARGET	0.76657	0.770388	0.765534	0.758602	0.769773	0.765663	0.763173
FOLD_80_BASE_PENALTY	2.65896	2.52053	2.62715	2.57759	2.53339	2.73908	2.61185
PCT_TARGET_BASES_1X	0.978378	0.980585	0.978068	0.977641	0.979909	0.977307	0.979238
PCT_TARGET_BASES_2X	0.970314	0.974374	0.970162	0.968836	0.972554	0.968607	0.971747
PCT_TARGET_BASES_10X	0.904993	0.924142	0.909823	0.900578	0.916909	0.902224	0.916938
PCT_TARGET_BASES_20X	0.801395	0.845312	0.818636	0.792799	0.83215	0.800699	0.836466
PCT_TARGET_BASES_30X	0.673627	0.744438	0.705237	0.661661	0.725257	0.67822	0.73483
PCT_TARGET_BASES_40X	0.544183	0.632262	0.586021	0.530369	0.609028	0.554277	0.623489
PCT_TARGET_BASES_50X	0.430247	0.524378	0.475962	0.415918	0.500019	0.444051	0.51661
PCT_TARGET_BASES_100X	0.124758	0.185763	0.153966	0.114152	0.171448	0.136547	0.183142
HS_LIBRARY_SIZE	0	0	0	0	0	0	0
HS_PENALTY_10X	0	0	0	0	0	0	0
HS_PENALTY_20X	0	0	0	0	0	0	0
HS_PENALTY_30X	0	0	0	0	0	0	0
HS_PENALTY_40X	0	0	0	0	0	0	0
HS_PENALTY_50X	0	0	0	0	0	0	0
HS_PENALTY_100X	0	0	0	0	0	0	0
AT_DROPOUT	8.97825	7.68819	10.2098	10.6797	9.2152	9.55367	10.3592
GC_DROPOUT	5.62923	5.60525	4.35947	3.91952	5.30564	5.61235	4.17348

HS METRICS (SureSelectXT_HS_Human_All_Exon_V8 BED):

MIN_MAP=15, MIN_BASE_QUAL_20, COVERAGE_CAP_1000									
BAIT_SET		BAIT							
	SureSelectXT_HS_Human_All_Exon_V8_BED								
NAME	MRS SHILPA V	MRS FAIZA FATHIMA	SAHANA	PRIYA	AYESHA	AMUL	LATHA	SUSILA	
GENOME_SIZE	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917
BAIT_TERRITORY	41623190	41623190	41623190	41623190	41623190	41623190	41623190	41623190	41623190
TARGET_TERRITORY	35131620	35131620	35131620	35131620	35131620	35131620	35131620	35131620	35131620
BAIT_DESIGN_EFFICIENCY	0.84404	0.84404	0.84404	0.84404	0.84404	0.84404	0.84404	0.84404	0.84404
TOTAL_READS	66656100	72449488	61992792	59354740	60037138	69100306	60912356	58561652	
PF_READS	66656100	72449488	61992792	59354740	60037138	69100306	60912356	58561652	
PF_UNIQUE_READS	66656100	72449488	61992792	59354740	60037138	69100306	60912356	58561652	
PCT_PF_READS	1	1	1	1	1	1	1	1	1
PCT_PF_UQ_READS	1	1	1	1	1	1	1	1	1
PF_UQ_READS_ALIGNED	66651232	72444696	61986880	59351333	60032953	69095822	60905466	58556021	
PCT_PF_UQ_READS_ALIGNED	0.999927	0.999934	0.999905	0.999943	0.99993	0.999935	0.999887	0.999904	
PF_BASES_ALIGNED	8689052862	9438018353	7887850461	7676464049	8037073568	9007241675	7969904084	7699926279	
PF_UQ_BASES_ALIGNED	8689052862	9438018353	7887850461	7676464049	8037073568	9007241675	7969904084	7699926279	
ON_BAIT_BASES	6161072791	6727996116	5474324815	5489077825	5564125162	6379790333	5567198550	5352202097	
NEAR_BAIT_BASES	1788558205	1917382718	1467371607	1522412824	1747735924	1846122563	1654441476	1637139369	
OFF_BAIT_BASES	739421866	792639519	946154039	664973400	725212482	781328779	748264058	710584813	
ON_TARGET_BASES	4110633334	4471909316	3692069670	3678246159	3650598985	4254267740	3707335778	3542796255	
PCT_SELECTED_BASES	0.914902	0.916016	0.880049	0.913375	0.909767	0.913255	0.906114	0.907715	
PCT_OFF_BAIT	0.0850981	0.0839837	0.119951	0.086625	0.0902334	0.0867445	0.0938862	0.0922846	
ON_BAIT_VS_SELECTED	0.775014	0.778219	0.788615	0.782869	0.760972	0.775572	0.770905	0.765766	
MEAN_BAIT_COVERAGE	148.02	161.641	131.521	131.875	133.678	153.275	133.752	128.587	
MEAN_TARGET_COVERAGE	117.007	127.29	105.092	104.699	103.912	121.095	105.527	100.844	
PCT_USABLE_BASES_ON_BAIT	0.678486	0.681861	0.66178	0.683888	0.663164	0.678072	0.667643	0.664255	
PCT_USABLE_BASES_ON_TARGET	0.452682	0.453214	0.446327	0.458275	0.435099	0.452162	0.4446	0.439692	
FOLD_ENRICHMENT	54.8083	55.102	53.6456	55.2714	53.5133	54.7491	53.9941	53.729	
ZERO_CVG_TARGETS_PCT	0.0152839	0.0155297	0.0161884	0.0148709	0.0153134	0.0154805	0.0153773	0.0154068	
PCT_EXC_DUPE	0	0	0	0	0	0	0	0	
PCT_EXC_MAPQ	0.0407308	0.0419605	0.0420658	0.0438151	0.0400569	0.0438667	0.0408363	0.0407464	
PCT_EXC_BASEQ	0.0105626	0.00979669	0.00982398	0.00846457	0.00975991	0.00988007	0.0104378	0.0122053	
PCT_EXC_OVERLAP	0.349733	0.354508	0.368645	0.360567	0.330462	0.349384	0.343801	0.336302	
PCT_EXC_OFF_TARGET	0.77733	0.773571	0.782206	0.776476	0.769774	0.776094	0.777532	0.777469	
FOLD_80_BASE_PENALTY	1.8001	1.79282	1.87665	1.90362	1.62363	1.83477	1.75878	1.80078	
PCT_TARGET_BASES_1X	0.982607	0.982452	0.981506	0.982614	0.982765	0.982512	0.982469	0.982367	
PCT_TARGET_BASES_2X	0.980923	0.980872	0.979621	0.980634	0.981302	0.980787	0.980773	0.980464	
PCT_TARGET_BASES_10X	0.970504	0.971081	0.966767	0.964765	0.973565	0.969604	0.970417	0.96799	
PCT_TARGET_BASES_20X	0.95437	0.956359	0.94556	0.940766	0.962465	0.952697	0.953156	0.947907	
PCT_TARGET_BASES_30X	0.932492	0.936812	0.916255	0.910379	0.944162	0.930735	0.928316	0.919986	
PCT_TARGET_BASES_40X	0.904044	0.912052	0.877098	0.871079	0.915359	0.902707	0.894187	0.882012	
PCT_TARGET_BASES_50X	0.866941	0.880638	0.826635	0.821418	0.873961	0.867245	0.848942	0.831671	
PCT_TARGET_BASES_100X	0.563214	0.617978	0.475584	0.47802	0.494435	0.580966	0.491157	0.455524	
HS_LIBRARY_SIZE	0	0	0	0	0	0	0	0	
HS_PENALTY_10X	0	0	0	0	0	0	0	0	
HS_PENALTY_20X	0	0	0	0	0	0	0	0	
HS_PENALTY_30X	0	0	0	0	0	0	0	0	
HS_PENALTY_40X	0	0	0	0	0	0	0	0	
HS_PENALTY_50X	0	0	0	0	0	0	0	0	
HS_PENALTY_100X	0	0	0	0	0	0	0	0	
AT_DROPOUT	0.260643	0.23186	0.34939	0.186917	0.162836	0.256858	0.233376	0.208519	
GC_DROPOUT	7.76424	8.53633	7.7885	9.99518	5.88693	8.48812	7.62471	8.06189	

STEP 5 : SECONDARY ANALYSIS (BAM AND VCF GENERATION):

- INPUT FILES : TRIMMED R1, R2 FASTQ FILES
- TOOL : SENTIEON
- TIME TAKEN: (5 HRS FOR 15 SAMPLES)
- DATA SIZE : 80 GB (FINAL OUTPUT)

SECONDARY ANALYSIS RESULT:

VCF AND BAM FILES

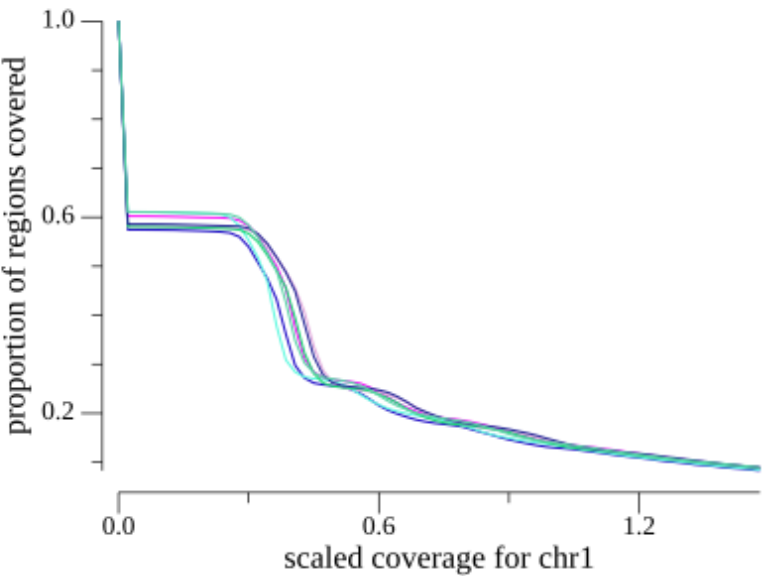
STEP 6 : INDEXCOVERAGE

- INPUT FILES : BAM FILES
- TOOL : INDEXCOV
- TIME TAKEN: (1 HRS)
- DATA SIZE : 10 GB (FINAL OUTPUT)

INDEXCOVERAGE SureSelectXT_Clinical_Research_Exome_V4 BAM (7 SAMPLES):

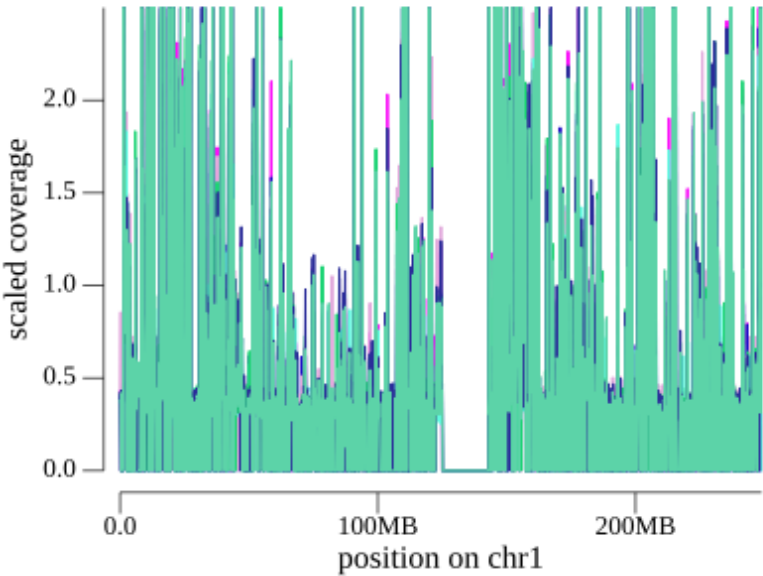
Coverage Plots [?](#)

Click each plot for an interactive view.



Depth Plots [?](#)

Click each plot for an interactive view.

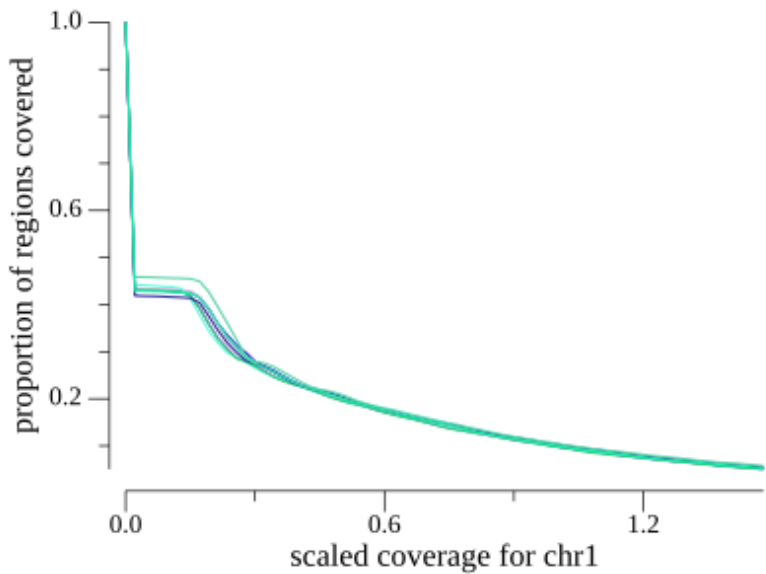


ALL CHROMOSOME COVERAGE STATISTICS AVAILABLE IN HTML LINK

INDEXCOVERAGE SureSelectXT_HS_Human_All_Exon_V8 BAM (8 SAMPLES) :

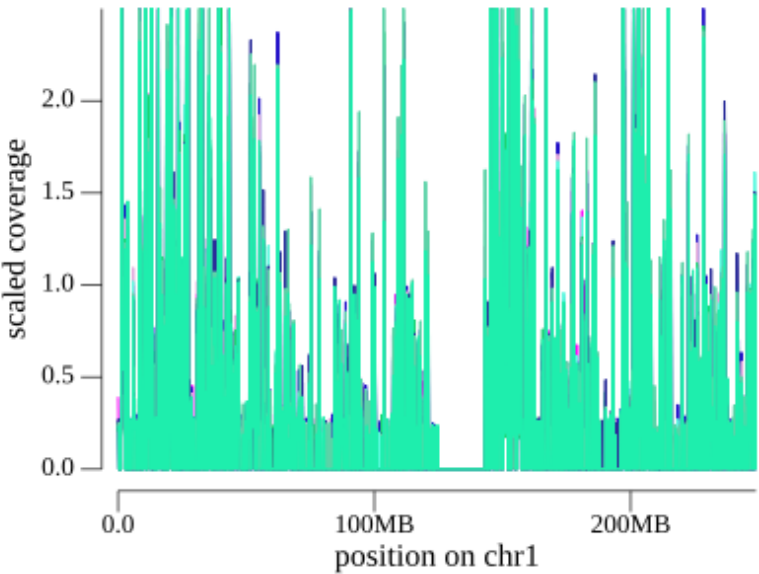
Coverage Plots [?](#)

Click each plot for an interactive view.



Depth Plots [?](#)

Click each plot for an interactive view.



ALL CHROMOSOME COVERAGE STATISTICS AVAILABLE IN HTML LINK

STEP 7 : VS CLINICAL SAMPLE AND COVERAGE STASTICS:

- INPUT FILES : VCF, BAM, BED FILES
- TOOL : VARSEQ
- TIME TAKEN: (3 DAYS 15 SAMPLES)
- DATA SIZE : 6 GB (FINAL OUTPUT)

RESULT: SAMPLE STASTICS (SureSelectXT_Clinical_Research_Exome_V4 BED 4 SAMPLES)

SAMPLE STASTICS									
	SUKANYA		VANISHREE		ASHIFA		BOGESH		
	SureSelectXT_Clinical_Research_Exome_V4	KAPPA BED (NUCLEOME)	SureSelectXT_Clinical_Research_Exome_V4	MEDGENOME BED	SureSelectXT_Clinical_Research_Exome_V4	MEDGENOME BED	SureSelectXT_Clinical_Research_Exome_V4	MEDGENOME BED	
Transition Count	286534	94441	317251	136549	278954	156335	294521	176102	
Transversion Count	144383	42815	158529	64208	138602	73589	147699	84446	
Ti/Tv	1.984541116	2.205792362	2.001217443	2.126666459	2.01262608	2.124434358	1.994062248	2.085380006	
Coding Transition Count	19098	22050	20454	21082	19606	20619	19900	21216	
Coding Transversion Count	6632	8285	7182	7803	6905	7063	7094	7899	
Coding Ti/Tv	2.879674306	2.661436331	2.847953216	2.701781366	2.839391745	2.690721649	2.805187482	2.685909609	
Singleton Count	445012	157834	491317	233128	431196	265795	456625	300773	
Het Count	67792	83893	85249	130456	73823	134751	78137	133438	
Het Ratio	0.152337465	0.531526794	0.173511195	0.559589582	0.171205206	0.506973419	0.1711118533	0.443650195	
Hom Count	377220	73941	406068	102672	357373	131044	378488	167335	
Hom Ratio	0.847778744	0.469246195	0.826604892	0.441664598	0.828910135	0.49415509	0.829055766	0.557486007	
Hemi Count	0	0	0	0	0	0	0	0	
Hemi Ratio	0	0	0	0	0	0	0	0	
SNV Het Count	64476	72447	81076	110230	70261	114153	74344	113587	
SNV Hom Count	366309	64668	394560	90215	347169	115423	367675	146532	
SNV Het/Hom Ratio	0.176015331	1.120291334	0.20548459	1.221858893	0.202382701	0.988996994	0.202200313	0.775168564	
Indel Het Count	3194	10926	4035	18902	19384	19384	3601	18625	
Indel Hom Count	10911	9273	11508	12457	10204	15621	10813	20803	
Indel Het/Hom Ratio	0.292732105	1.178259463	0.350625652	1.517379786	0.337318699	1.240893669	0.333025062	0.895303562	
Var Count	445012	157834	491317	233128	431196	265795	456625	300773	
Coding Var Count	26266	31671	28190	29991	27046	29360	27538	30229	
Ref Count	0	0	0	0	0	0	0	0	
Ref Ratio	0	0	0	0	0	0	0	0	
Called Genotypes	445012	157834	491317	233128	431196	265795	456625	300773	
Call Rate	1	1	1	1	1	1	1	1	
SNP Count	430873	137209	475732	200653	417514	229808	442153	260405	
MNP Count	0	0	0	0	0	0	0	0	
Ins Count	5816	6436	9137	14465	5661	16151	5963	17895	
Del Count	8310	10788	9137	17620	8009	19513	8490	22171	
DelIns Count	0	0	0	0	0	0	0	0	
Complex Count	13	131	12	370	12	323	19	302	
Chrom X Het Ratio	0.033976655	0.48722467	0.165675983	0.541043862	0.035193133	0.405489846	0.033367382	0.119222097	
Inferred Gender	Male	Female	Male	Female	Male	Female	Male	Male	
Multiallelic Site Count	61	260	69	662	607	607	613	613	
Multiallelic Het Count	61	260	69	662	607	607	613	613	
Coding Multiallelic Site Count	14	34	47	47	13	33	19	46	
Coding Multiallelic Het Count	14	34	47	47	13	33	19	46	
Q20 Values	6.472733 G (97.404035%)	4.902472 G (98.465368%)	7.445040 G (98.06057%)	8.866446 G (97.207834%)	6.733793 G (97.459231%)	9.123243 G (97.4459289%)	6.348655 G (97.676289%)	8.874134 G (97.511701%)	
Q30 Values	6.114249 G (92.009432%)	4.757541 G (95.54452%)	7.130416 G (93.921829%)	8.427033 G (92.390300%)	6.363815 G (92.104470%)	8.701137 G (92.729383%)	6.027122 G (92.729383%)	8.469365 G (93.063977%)	
TOTAL READS GENEREATED	7.0 GB	6.0 GB	8.0 GB	6.0 GB	7.3 GB	7.1 GB	6.7 GB	7.1 GB	

RESULT: SAMPLE STASTICS (SureSelectXT_Clinical_Research_Exome_V4 BED 3 SAMPLES)

SAMPLE STASTICS						
	ARADHAYA		PRANEETH		IDHAYAKANI	
	SureSelectXT_Clinical_Research_Exome_V4	MEDGENOME BED	SureSelectXT_Clinical_Research_Exome_V4	MEDGENOME BED	SureSelectXT_Clinical_Research_Exome_V4	
Transition Count	308768	198396	279372	184626	307917	
Transversion Count	154625	96168	139753	88667	154657	
Ti/Tv	1.996882781	2.063014724	1.999041165	2.082240292	1.990967108	
Coding Transition Count	19898	21296	18704	21012	19990	
Coding Transversion Count	7066	7919	6709	7798	7157	
Coding Ti/Tv	2.816020379	2.689228438	2.787896855	2.694537061	2.793069722	
Singleton Count	478638	343868	432169	316895	477295	
Het Count	79822	161404	65277	140150	79847	
Het Ratio	0.16676904	0.469377784	0.151045077	0.442260055	0.167290669	
Hom Count	398816	182464	366892	176745	397448	
Hom Ratio	0.833366767	0.532191545	0.849086558	0.55906132	0.832815768	
Hemi Count	0	0	0	0	0	
Hemi Ratio	0	0	0	0	0	
SNV Het Count	75908	134877	62083	118081	76101	
SNV Hom Count	387332	159213	356886	154870	386344	
SNV Het/Hom Ratio	0.195976578	0.84714816	0.17395751	0.762452379	0.196977305	
Indel Het Count	3758	24499	3060	20571	3624	
Indel Hom Count	11484	23251	10006	21875	11104	
Indel Het/Hom Ratio	0.327237896	1.053675111	0.30581651	0.940388571	0.326368876	
Var Count	478638	343868	432169	316895	477295	
Coding Var Count	27533	30382	25902	29982	27725	
Ref Count	0	0	0	0	0	
Ref Ratio	0	0	0	0	0	
Called Genotypes	478638	343868	432169	316895	477295	
Call Rate	1	1	1	1	1	
SNP Count	463342	294406	419073	273179	462531	
MNP Count	0	0	0	0	0	
Ins Count	6390	21751	5406	19336	5986	
Del Count	8888	27149	7679	23972	8767	
Dellns Count	0	0	0	0	0	
Complex Count	18	562	11	408	11	
Chrom X Het Ratio	0.039776463	0.383431612	0.035906433	0.091305463	0.034439787	
Inferred Gender	Male	Female	Male	Male	Male	
Multiallelic Site Count	78	1014	67	749	61	
Multiallelic Het Count	78	1014	67	749	61	
Coding Multiallelic Site Count	10	51	15	33	7	
Coding Multiallelic Het Count	10	51	15	33	7	
Q20 Values	7.259854 G (97.899841%)	13.289725 G (98.337132%)	6.482534 G (97.429012%)	12.941381 G (97.114176%)	7.382220 G (97.525439%)	
Q30 Values	6.924786 G (93.381410%)	12.827643 G (94.917965%)	6.125810 G (92.067650%)	12.289707 G (92.223907%)	6.989570 G (92.338196%)	
TOTAL READS GENEREATED	7.8 GB	6.2 GB	7.0 GB	6.3 GB	8.0 GB	

RESULT: SAMPLE STASTICS (SureSelectXT_HS_Human_All_Exon_V8 BED 8 SAMPLES):

SAMPLE STASTICS									
	MRS SHILPA V	MRS FAIZA FATHIMA	SAHANA	PRIYA	AYESHA	AMUL	LATHA	SUSILA	
	SureSelectXT_HS_Human_All_Exon_V8	SureSelectXT_HS_Human_All_Exon_V8	SureSelectXT_HS_Human_All_Exon_V8	SureSelectXT_HS_Human_All_Exon_V8	SureSelectXT_HS_Human_All_Exon_V8	SureSelectXT_HS_Human_All_Exon_V8	SureSelectXT_HS_Human_All_Exon_V8	SureSelectXT_HS_Human_All_Exon_V8	
Transition Count	119084	124265	147407	112450	116020	123218	119830	114552	
Transversion Count	53750	57385	114095	54446	53126	59298	54438	51886	
Ti/Tv	2.215516279	2.165461357	1.29196722	2.065349153	2.183864774	2.077945293	2.201219736	2.207763173	
Coding Transition Count	19583	19608	19334	19080	19445	19157	19223	19315	
Coding Transversion Count	6912	6896	6955	6774	6804	6652	6652	6693	
Coding Ti/Tv	2.833188657	2.843387471	2.779870597	2.894417476	2.870534396	2.815549677	2.889807577	2.885850889	
Singleton Count	179928	188957	268269	173020	176748	189710	181338	173380	
Het Count	48505	49265	46346	42872	48662	43730	47355	47322	
Het Ratio	0.269580054	0.260720693	0.172759432	0.247786383	0.275318533	0.230509725	0.261142176	0.272938055	
Hom Count	131423	139692	221923	130148	128086	145980	133983	126058	
Hom Ratio	0.730606731	0.739451494	0.827348509	0.752426707	0.724907042	0.769672844	0.739024915	0.727242308	
Hemi Count	0	0	0	0	0	0	0	0	
Hemi Ratio	0	0	0	0	0	0	0	0	
SNV Het Count	45691	46311	43920	40451	45592	41109	44649	44575	
SNV Hom Count	127077	135267	217522	126361	123467	141323	129529	121788	
SNV Het/Hom Ratio	0.359553656	0.342367318	0.201910611	0.320122506	0.369264662	0.290886834	0.344702731	0.366004861	
Indel Het Count	2722	2866	2356	2323	2960	2531	2624	2661	
Indel Hom Count	4346	4425	4401	3787	4657	4657	4454	4270	
Indel Het/Hom Ratio	0.626323056	0.647683616	0.535332879	0.613414312	0.640631349	0.543482929	0.589133363	0.623185012	
Var Count	179928	188957	268269	173020	176748	189710	181338	173380	
Coding Var Count	27058	27104	26871	26225	26821	26554	26439	26563	
Ref Count	0	0	0	0	0	0	0	0	
Ref Ratio	0	0	0	0	0	0	0	0	
Called Genotypes	179928	188957	268269	173020	176748	189710	181338	173380	
Call Rate	1	1	1	1	1	1	1	1	
SNP Count	172812	181626	261482	166868	169117	182488	174238	166413	
MNP Count	0	0	0	0	0	0	0	0	
Ins Count	2931	2988	2845	2574	3209	3005	2947	2964	
Del Count	4168	4327	3933	3563	4401	4204	4143	3988	
DelIns Count	0	0	0	0	0	0	0	0	
Complex Count	17	16	9	15	21	13	10	15	
Chrom X Het Ratio	0.217699115	0.229550321	0.12779862	0.050198628	0.215330189	0.208246564	0.217504333	0.202519832	
Inferred Gender	Male	Male	Male	Male	Male	Male	Male	Male	
Multiallelic Site Count	46	44	35	49	55	45	41	43	
Multiallelic Het Count	46	44	35	49	55	45	41	43	
Coding Multiallelic Site Count	12	8	7	11	12	13	14	13	
Coding Multiallelic Het Count	12	8	7	11	12	13	14	13	
Q20 Values	8.872567 G (97.708796%)	9.655377 G (97.854230%)	8.094197 G (97.849146%)	7.878809 G (98.162681%)	8.204869 G (97.790311%)	9.205526 G (97.840401%)	8.146192 G (97.692705%)	7.840713 G (97.310113%)	
Q30 Values	8.429104 G (92.825182%)	9.202123 G (93.260645%)	7.711394 G (93.221516%)	7.558015 G (94.165878%)	7.813154 G (93.121620%)	8.768535 G (93.195866%)	7.738131 G (92.799055%)	7.394524 G (91.772523%)	
TOTAL READS GENEREATED	10 GB	10.8 GB	9.3 GB	8.9 GB	9.0 GB	10.3 GB	9.1 GB	8.8 GB	

RESULT:COVERAGE STASTICS(SureSelectXT_Clinical_Research_Exome_V4 BED 4 SAMPLES)

	COVERAGE STASTICS									
	SUKANYA		VANISHREE		ASHIFA		BOGESH			
	SureSelectXT_Clinical_Research_Exome_V4	KAPPA BED (NUCLEOME)	SureSelectXT_Clinical_Research_Exome_V4	MEDGENOME BED	SureSelectXT_Clinical_Research_Exome_V4	MEDGENOME BED	SureSelectXT_Clinical_Research_Exome_V4	MEDGENOME BED		
Sample Mean Depth	54.8479		53.2456		101.122		60.3764		54.068	108.84
Sample Mean Forward Depth	27.4326		26.5937		50.5461		30.1957		52.7004	54.4149
Sample Mean Reverse Depth	27.4152		26.6517		50.5759		30.1806		52.7242	54.425
Sample Mean Filtered Depth	2.42031		2.81038		3.54216		2.46062		3.04444	3.38938
Sample % 1x	95.4366		97.9892		97.5162		95.4123		96.3786	97.8192
Sample % 5x	91.4848		97.5565		92.2023		91.5882		94.533	96.715
Sample % 20x			78.5856		82.0365		80.1376		90.9992	94.2001
Sample % 50x			44.9046		55.5184		52.9523		86.6103	87.2489
Sample % 100x			14.1201		2.26929		19.8186		55.6424	62.045
Sample % 500x			0.0639965		0.00854059		0.0548215		0.010162	0.0115619
Sample Errors			?		?		?		?	?
Q20 Values	6.472733 G (97.404035%)		4.902472 G (98.465368%)		7.445040 G (98.066057%)		8.866446 G (97.207834%)		6.733793 G (97.459231%)	9.123243 G (97.445939%)
Q30 Values	6.114249 G (92.009432%)		4.757541 G (95.554452%)		7.130416 G (93.921825%)		8.427033 G (92.390300%)		6.363815 G (92.104470%)	8.701137 G (92.937400%)
TOTAL READS GENERATED	7.0 GB		6.0 GB		8.0 GB		6.0 GB		7.3 GB	7.1 GB

RESULT:COVERAGE STASTICS(SureSelectXT_Clinical_Research_Exome_V4 BED 3 SAMPLES)

COVERAGE STASTICS							
	ARADHAYA		PRANEETH		IDHAYAKANI		
	SureSelectXT_Clinical_Research_Exome_V4	MEDGENOME BED	SureSelectXT_Clinical_Research_Exome_V4	MEDGENOME BED	SureSelectXT_Clinical_Research_Exome_V4		
Sample Mean Depth	61.9484	166.592	56.7719	150.048	64.9357		
Sample Mean Forward Depth	30.9815	83.276	28.3958	75.0128	32.4768		
Sample Mean Reverse Depth	30.9668	83.3154	28.376	75.0354	32.4588		
Sample Mean Filtered Depth	2.66306	4.97229	2.39966	4.50432	2.73858		
Sample % 1x	95.6722	96.8209	95.3463	97.6673	95.5719		
Sample % 5x	91.9406	95.3983	91.2727	96.5307	91.899		
Sample % 20x	81.0405	93.8927	78.5251	94.4677	81.5976		
Sample % 50x	50.9348	91.242	46.2998	89.8955	53.2462		
Sample % 100x	18.5005	84.0543	15.5093	77.2551	20.4294		
Sample % 500x	0.069064	0.0439869	0.0552437	0.0309434	0.0651165		
Sample Errors	?	?	?	?	?		
Q20 Values	7.259854 G (97.899841%)	13.289725 G (98.337132%)	6.482534 G (97.429012%)	12.941381 G (97.114176%)	7.382220 G (97.525439%)		
Q30 Values	6.924786 G (93.381410%)	12.827643 G (94.917965%)	6.125810 G (92.067650%)	12.289707 G (92.223907%)	6.989570 G (92.338196%)		
TOTAL READS GENEREATED	7.8 GB	6.2 GB	7.0 GB	6.3 GB	8.0 GB		

RESULT: COVERAGE STASTICS (SureSelectXT_HS_Human_All_Exon_V8 BED 8 SAMPLES):

COVERAGE STASTICS									
	MRS SHILPA V	MRS FAIZA FATHIM A	SAHANA	PRIYA	AYESHA	AMUL	LATHA	SUSILA	
	SureSelectXT_HS_Human_All_Exon_V8	SureSelectXT_HS_Human_All_Exon_V8	SureSelectXT_HS_Human_All_Exon_V8	SureSelectXT_HS_Human_All_Exon_V8	SureSelectXT_HS_Human_All_Exon_V8	SureSelectXT_HS_Human_All_Exon_V8	SureSelectXT_HS_Human_All_Exon_V8	SureSelectXT_HS_Human_All_Exon_V8	
Sample Mean Depth	102.679	111.153	91.839	90.4056	90.0368	105.579	93.1651	89.385	
Sample Mean Forward Depth	51.349	55.5873	45.9272	45.21	45.0268	52.7997	46.5917	44.7007	
Sample Mean Reverse Depth	51.3301	55.5652	45.9117	45.1954	45.0099	52.7793	46.5733	44.6842	
Sample Mean Filtered Depth	4.65521	5.35631	4.33668	4.63385	4.19265	5.35455	4.27171	4.10757	
Sample % 1x	97.4758	97.4437	97.3015	97.44	97.4752	97.4545	97.4389	97.4517	
Sample % 5x	96.558	96.5581	96.2772	96.298	96.6496	96.4974	96.5072	96.451	
Sample % 20x	93.8082	94.0254	92.7741	92.2824	94.5656	93.6021	93.676	93.1554	
Sample % 50x	83.2057	84.9303	78.2859	77.6662	83.2046	83.2136	81.0677	79.2224	
Sample % 100x	47.3504	53.1619	38.5088	37.9539	37.9257	49.0875	40.0793	36.8631	
Sample % 500x	0.0784412	0.0872025	0.0494937	0.0551538	0.0308743	0.0868929	0.0431967	0.0383245	
Sample Errors	?	?	?	?	?	?	?	?	
Q20 Values	8.872567 G (97.708796%)	9.655377 G (97.854230%)	8.094197 G (97.849146%)	7.878809 G (98.162681%)	8.204869 G (97.790311%)	9.205526 G (97.840401%)	8.146192 G (97.692705%)	7.840713 G (97.310113%)	
Q30 Values	8.429104 G (92.825182%)	9.202123 G (93.260645%)	7.711394 G (93.221516%)	7.558015 G (94.165878%)	7.813154 G (93.121620%)	8.768535 G (93.195866%)	7.738131 G (92.799055%)	7.394524 G (91.772523%)	
TOTAL READS GENEREATED	10 GB	10.8 GB	9.3 GB	8.9 GB	9.0 GB	10.3 GB	9.1 GB	8.8 GB	

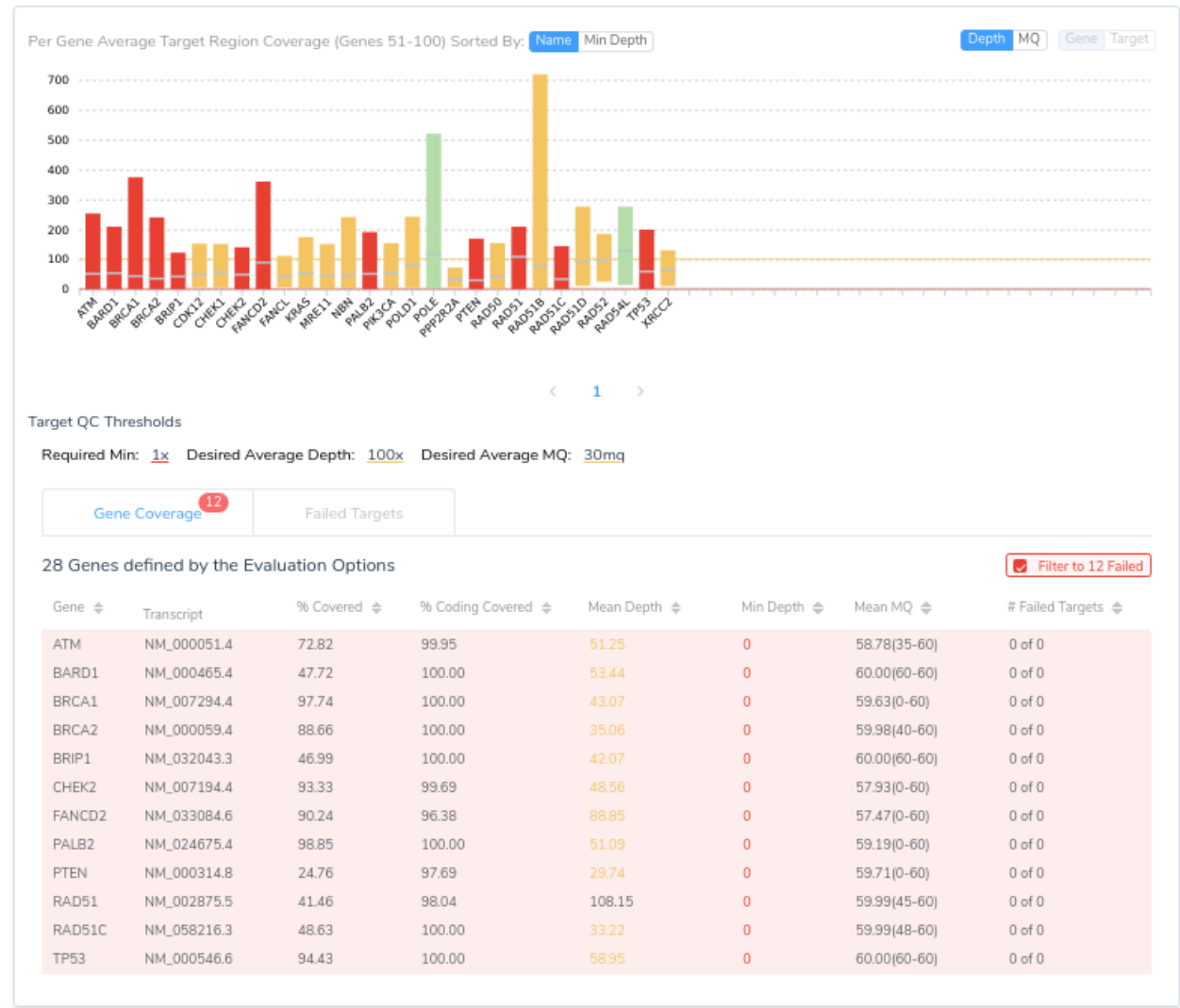
CANCER GENE DEPTH COVERAGE: (ASHIFA_4)

GENE LIST :

ATM, BARD1, BRCA1, BRCA2, BRIP1,
CDK12, CHEK2, FANCD2, MRE11, NBN,
PALB2, PPP2R2A, RAD51B, RAD54L,
TP53, CHEK1, FANCL, RAD50, RAD51,
RAD51C, RAD51D, RAD52, XRCC2,
KRAS, PIK3CA, POLD1, POLE, PTEN

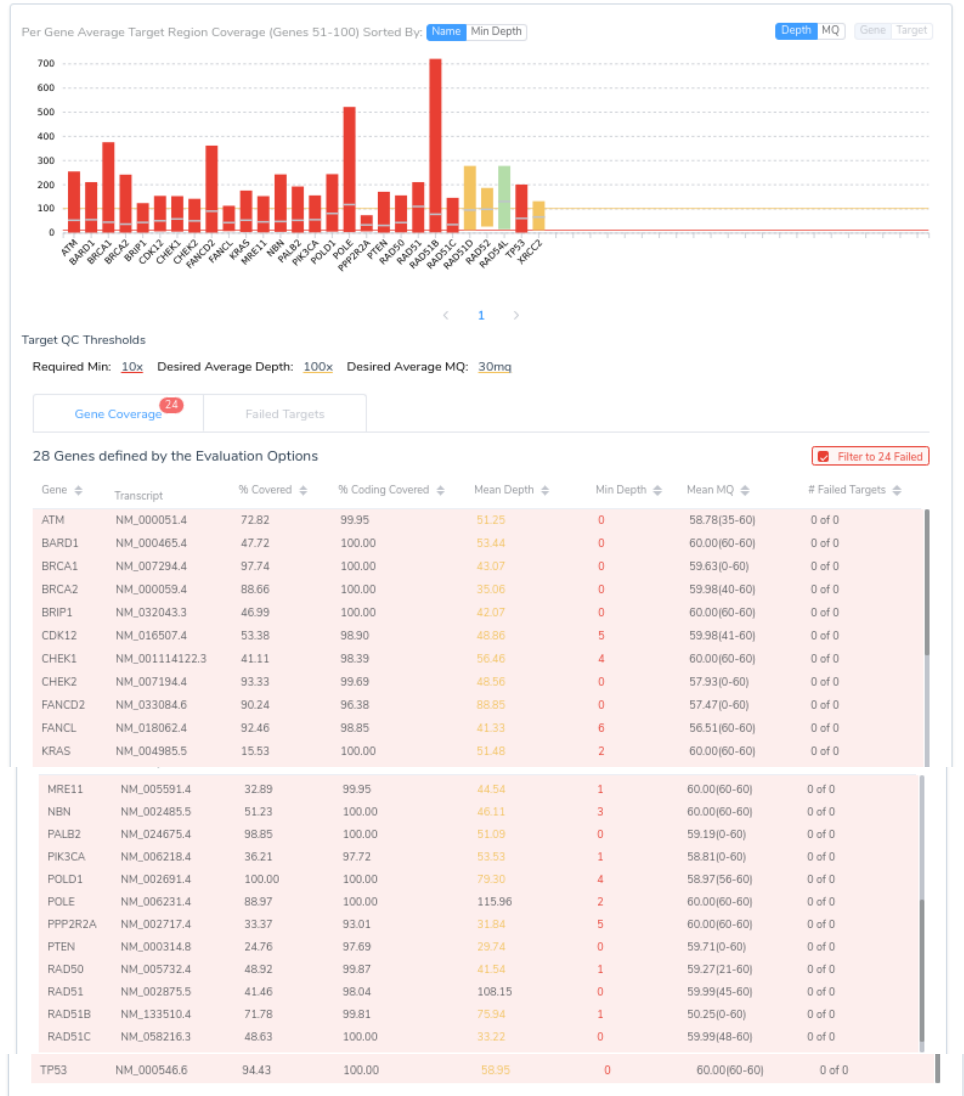
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Reported Failed Targets



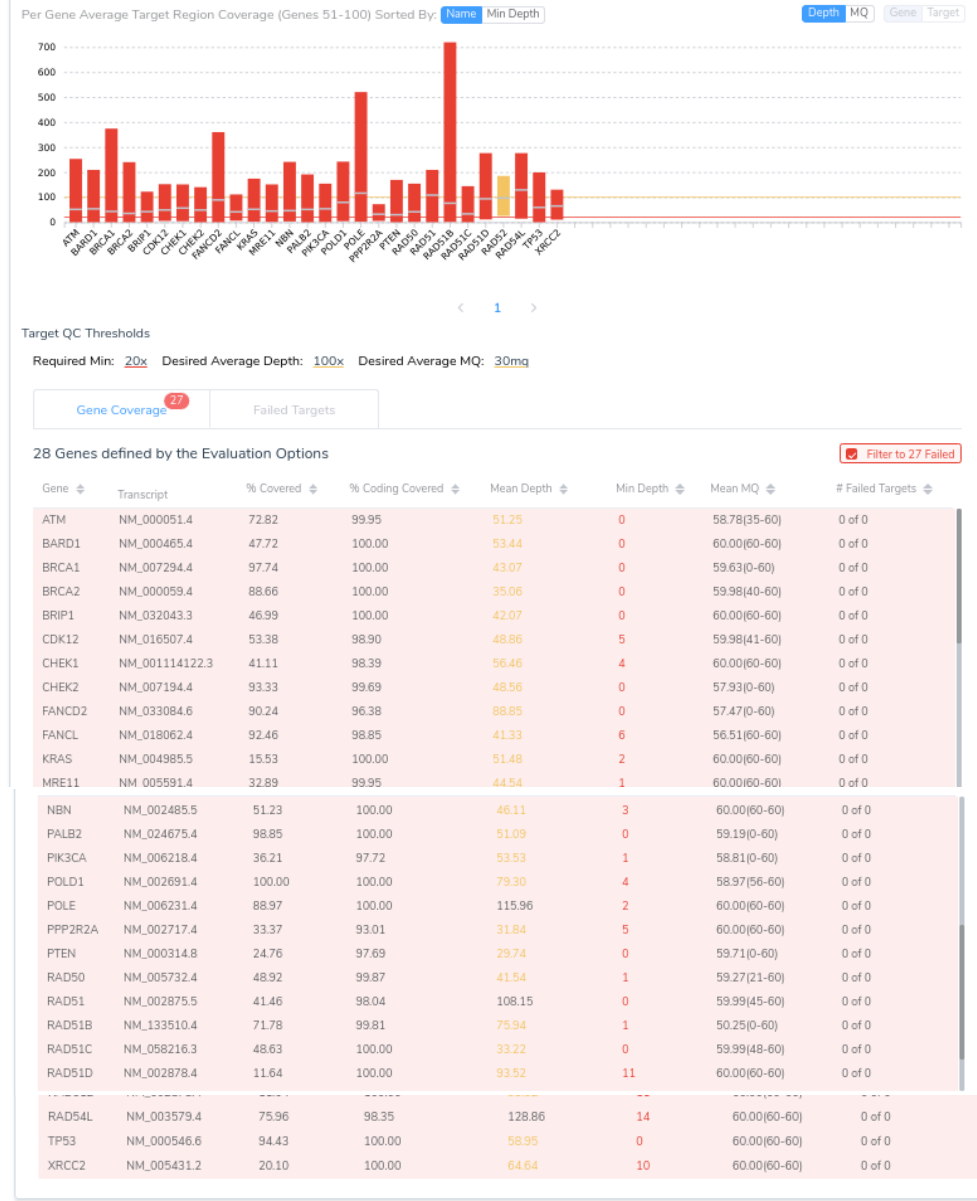
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Reported Failed Targets



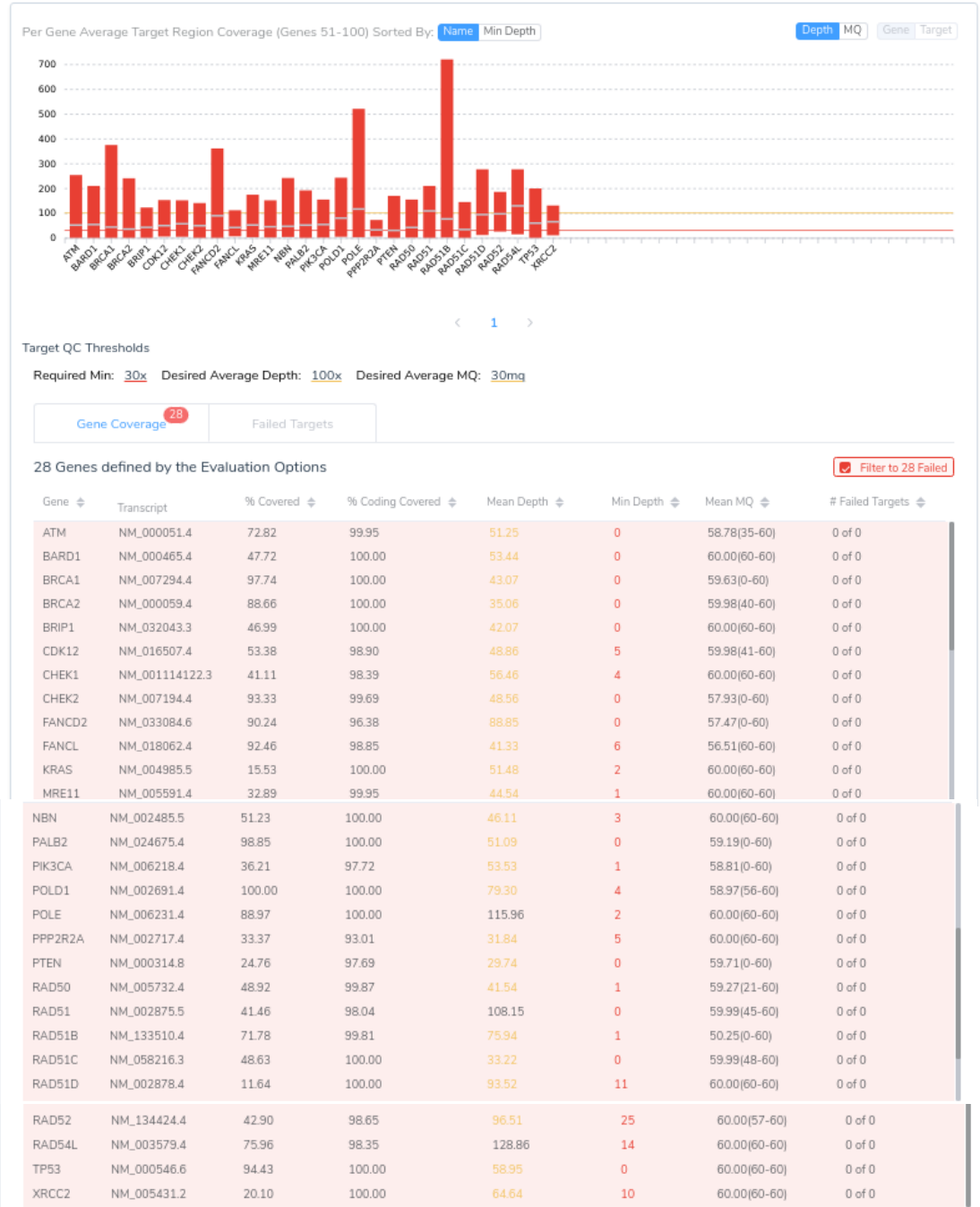
20X :

Reported Failed Targets



30X :

Reported Failed Targets



VARIANT DEPTH COMPARISSION :